SURAJ YADAV
SQL Portfolio Project

Tool: MySQL 8.0

PIZZA SALES DATA ANALYSIS

- END-TO-END BUSINESS PERFORMANCE & SQL INSIGHTS



SURAJ YADAV
SQL Portfolio Project



PROJECT OVERVIEW

EXECUTIVE SUMMARY & DATA SCHEMA

Objective: Analyzed 48,000+ pizza sales records using MySQL to evaluate revenue drivers, peak operational hours, and menu profitability.

Database Architecture (4 Relational Tables):

- orders: Order ID, date, and order timestamp
- order_details: Transaction line items, pizza ID, and quantity.
- pizzas: Pizza sizes (S, M, L, XL), unit price, and category mapping.
- pizza_types: Pizza names, recipe categories, and ingredient breakdown.

SURAJ YADAV
SQL Portfolio Project

RETRIEVE THE TOTAL NUMBER OF ORDERS PLACED

```sql
SELECT
    COUNT(order_id) AS total_order
FROM
    orders;
```



Business Insights

- Volume Baseline: Processed 21,350 unique orders over the year, averaging around 58–60 orders daily
- Operational Impact: Gives the kitchen and procurement teams a dependable benchmark to forecast daily dough prep, fresh ingredient purchases, and peak-hour staffing.

SURAJ YADAV
SQL Portfolio Project

Calculating cumulative gross revenue generated across all pizza sales and transactions

```
SELECT
    ROUND(SUM(order_details.quantity * pizzas.price),
            2) AS total_revenue
FROM
    pizzas
        JOIN
    order_details ON pizzas.pizza_id = order_details.pizza_id;
```

total_revenue
817860.05

Business Insights

• Gross Revenue: The business generated $817,860.05 in total gross sales across 21,350 completed orders.

• Average Order Value (AOV): Translates to an AOV of ~$38.30 per customer transaction, serving as a core benchmark for future combo bundles and upsell initiatives.

SURAJ YADAV
SQL Portfolio Project

IDENTIFY THE HIGHEST-PRICED PIZZA



```sql
SELECT
    pizza_types.name, pizzas.price
FROM
    pizzas
        JOIN
    pizza_types ON pizzas.pizza_type_id = pizza_types.pizza_type_id
ORDER BY pizzas.price DESC
LIMIT 1;
```

	name	price
▶	The Greek Pizza	35.95

Business Insights

- Premium SKU: The Greek Pizza (XL) represents the peak price point on the menu at $35.95
- Strategic Value: High-ticket anchor items drive overall ticket sizes and test price sensitivity without lowering order volume

SURAJ YADAV

SQL Portfolio Project

MOST COMMON PIZZA SIZE ORDERED

```sql
SELECT
    pizzas.size,
    COUNT(order_details.order_details_id) AS order_count
FROM
    order_details
JOIN
    pizzas ON order_details.pizza_id = pizzas.pizza_id
GROUP BY
    pizzas.size
ORDER BY
    order_count DESC;
```

size	order_count
L	18526
M	15385
S	14137
XL	544
XXL	28



Business Insights

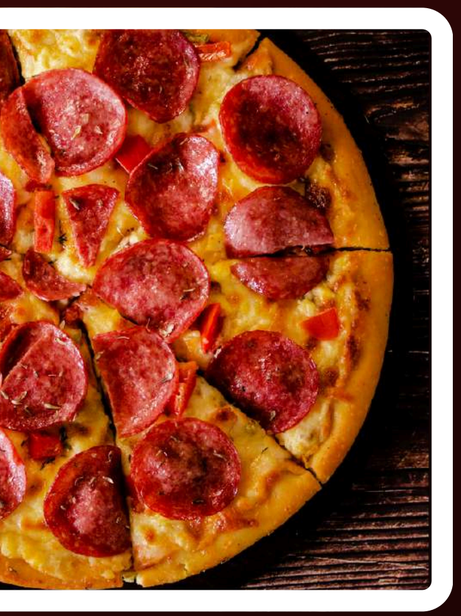
• Volume Driver: Large (L) size leads customer preference with 18,526 orders, followed by Medium (15,385) and Small (14,137).

• Supply Chain & Packaging: High demand for Large pizzas means procurement must prioritize Large pizza boxes, dough portions, and oven tray allocation to avoid bottlenecks during peak hours.

SURAJ YADAV

SQL Portfolio Project

LIST THE TOP 5 MOST-ORDERED PIZZA TYPES ALONG WITH THEIR QUANTITIES



```sql
SELECT
    pizza_types.name,
    SUM(order_details.quantity) AS total_quantity
FROM
    pizza_types
        JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
        JOIN
    order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.name
ORDER BY total_quantity DESC
LIMIT 5;
```

name	total_quantity
The Classic Deluxe Pizza	2453
The Barbecue Chicken Pizza	2432
The Hawaiian Pizza	2422
The Pepperoni Pizza	2418
The Thai Chicken Pizza	2371

Business Insights

• Core Revenue Drivers: The Classic Deluxe and Barbecue Chicken dominate customer orders, proving high baseline customer loyalty for classic and poultry recipes.

• Supply Chain Priority: Kitchen procurement must secure bulk inventory contracts for high-turnover ingredients (chicken, bacon, pepperoni, mozzarella) to prevent stockouts on top-selling SKUs.

JOIN THE NECESSARY TABLES TO FIND THE TOTAL QUANTITY OF EACH PIZZA CATEGORY ORDERED



```sql
SELECT
    pizza_types.category,
    SUM(order_details.quantity) AS total_quantity
FROM
    pizza_types
        JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
        JOIN
    order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.category
ORDER BY total_quantity DESC;
```

	category	total_quantity
▶	Classic	14888
	Supreme	11987
	Veggie	11649
	Chicken	11050

Business Insights

• Volume Leader: The Classic category accounts for 30% of total volume (14,888 units), driven by everyday staple favorites like Pepperoni and Hawaiian.

• Balanced Demand: Supreme (11,987), Veggie (11,649), and Chicken (11,050) show balanced, steady volume, ensuring no single recipe family creates operational fragility.

SURAJ YADAV

SQL Portfolio Project

Determine the distribution of orders by the hour of the day





```
SELECT
    HOUR(order_time) AS order_hour,
    COUNT(order_id) AS total_orders
FROM
    orders
GROUP BY HOUR(order_time)
ORDER BY total_orders DESC;
```

order_hour	total_orders
12	2520
13	2455
18	2399
17	2336
19	2009
16	1920
20	1642
14	1472
15	1468
11	1231
21	1198
22	663
23	28
10	8
9	1

Business Insights

• Dual Peak Periods: Order volume spikes sharply during Lunch (12 PM – 1 PM) and Dinner (6 PM – 7 PM), accounting for the majority of daily ticket volume.

• Labor Optimization: Management should implement staggered shift scheduling, deploying peak kitchen staff and delivery personnel during these windows while scheduling prep and cleaning during mid-afternoon lulls (3 PM – 5 PM).

SURAJ YADAV
SQL Portfolio Project

Find the category-wise distribution of pizzas

```sql
SELECT
    category, COUNT(name) AS total_pizza_types
FROM
    pizza_types
GROUP BY category
ORDER BY total_pizza_types DESC;
```

	category	total_pizza_types
▶	Supreme	9
	Veggie	9
	Classic	8
	Chicken	6

Business Insights

Menu Balance: The catalog maintains an even spread across Supreme (9), Veggie (9), and Classic (8), offering broad choices across diverse flavor profiles.

Product Development Opportunity: Chicken currently holds the smallest SKU count (6 items); adding 1–2 chicken variants could capture additional high-margin poultry demand.

SURAJ YADAV

SQL Portfolio Project

GROUP THE ORDERS BY DATE AND CALCULATE THE AVERAGE NUMBER OF PIZZAS ORDERED PER DAY

```sql
SELECT
    ROUND(AVG(daily_quantity), 0) AS avg_pizzas_per_day
FROM (
    SELECT
        orders.order_date,
        SUM(order_details.quantity) AS daily_quantity
    FROM
        orders
    JOIN
        order_details ON orders.order_id = order_details.order_id
    GROUP BY
        orders.order_date
) AS daily_orders;
```

	avg_pizzas_per_day
▶	138

Business Insights

- Daily Kitchen Target: The restaurant averages 138 pizzas per day, giving line cooks and shift leads an exact daily production baseline.
- Waste Reduction: Kitchen staff can pre-batch approximately 135–140 dough portions each morning, reducing rush-hour prep lag while preventing end-of-day spoilage.

SURAJ YADAV
SQL Portfolio Project

DETERMINE THE TOP 3 MOST-ORDERED PIZZA TYPES BASED ON REVENUE



```sql
SELECT
    pizza_types.name,
    ROUND(SUM(order_details.quantity * pizzas.price),
            2) AS total_revenue
FROM
    pizza_types
        JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
        JOIN
    order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.name
ORDER BY total_revenue DESC
```

name	total_revenue
The Thai Chicken Pizza	43434.25
The Barbecue Chicken Pizza	42768
The California Chicken Pizza	41409.5

Business Insights

• High-Margin Dominance: Chicken-based recipes dominate the top 3 revenue spots, proving that customers readily pay premium prices for specialty poultry pizzas.

• Strategic Promotion: Focus digital ads and high-margin upsell banners on these 3 SK' maximize gross contribution per mark dollar.

SURAJ YADAV
SQL Portfolio Project

Percentage Contribution of Each Pizza Category to Total Revenue

```sql
SELECT
    pizza_types.category,
    ROUND(SUM(order_details.quantity * pizzas.price) / (SELECT
                    ROUND(SUM(order_details.quantity * pizzas.price),
                                2)
                FROM
                    order_details
                        JOIN
                    pizzas ON pizzas.pizza_id = order_details.pizza_id) * 100,
            2) AS revenue_percentage
FROM
    pizza_types
        JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
        JOIN
    order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.category
ORDER BY revenue_percentage DESC;
```

category	revenue_percentage
Classic	26.91
Supreme	25.46
Chicken	23.96
Veggie	23.68

Business Insights

- Balanced Revenue Stream: All four categories contribute nearly evenly (~24%–27%), demonstrating healthy revenue diversification with no dangerous reliance on a single category.
- Classic Margin Value: Classic leads total revenue at ~26.9%, serving as the core bread-and-butter offering that anchors the restaurant's cash flow.

SURAJ YADAV
SQL Portfolio Project

Cumulative Revenue Generated Over Time

```sql
SELECT
    order_date,
    ROUND(SUM(revenue) OVER(ORDER BY order_date), 2) AS cum_revenue
FROM (
    SELECT
        orders.order_date,
        SUM(order_details.quantity * pizzas.price) AS revenue
    FROM
        order_details
    JOIN
        pizzas ON order_details.pizza_id = pizzas.pizza_id
    JOIN
        orders ON orders.order_id = order_details.order_id
    GROUP BY
        orders.order_date
) AS daily_sales
limit 5;
```

order_date	cum_revenue
2015-01-01	2713.85
2015-01-02	5445.75
2015-01-03	8108.15
2015-01-04	9863.6
2015-01-05	11929.55

Business Insights

- Predictable Trajectory: The cumulative revenue curve grows at a steady, linear rate across all 12 months, showing no major seasonal slump.
- Financial Forecasting: Steady daily run-rates make it easy for finance and operations to project quarterly cash flow and manage vendor payment cycles reliably.

SURAJ YADAV

SQL Portfolio Project

Top 3 Most-Ordered Pizzas by Revenue for Each Category

```sql
SELECT category, name, revenue
FROM (
    SELECT
        pizza_types.category,
        pizza_types.name,
        ROUND(SUM(order_details.quantity * pizzas.price), 2) AS revenue,
        DENSE_RANK() OVER(
            PARTITION BY pizza_types.category
            ORDER BY SUM(order_details.quantity * pizzas.price) DESC
        ) AS ranking
    FROM pizza_types
    JOIN pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
    JOIN order_details ON order_details.pizza_id = pizzas.pizza_id
    GROUP BY pizza_types.category, pizza_types.name
) AS ranked_pizzas
WHERE ranking <= 3;
```

category	name	revenue
Chicken	The Thai Chicken Pizza	43434.25
Chicken	The Barbecue Chicken Pizza	42768
Chicken	The California Chicken Pizza	41409.5
Classic	The Classic Deluxe Pizza	38180.5
Classic	The Hawaiian Pizza	32273.25
Classic	The Pepperoni Pizza	30161.75
Supreme	The Spicy Italian Pizza	34831.25
Supreme	The Italian Supreme Pizza	33476.75
Supreme	The Sicilian Pizza	30940.5
Veggie	The Four Cheese Pizza	32265.7
Veggie	The Mexicana Pizza	26780.75
Veggie	The Five Cheese Pizza	26066.5



Business Insights

• Hero Products: Across every category, the top 2–3 "Hero SKUs" account for over 60% of segment revenue, led by Thai Chicken ($43.4K) and BBQ Chicken ($42.8K).

• Menu Engineering: Shift marketing spend away from low-traction items and focus promotional bundles and digital combos around these proven revenue drivers.

SURAJ YADAV
SQL Portfolio Project

THANK YOU

FOR YOUR TIME & ATTENTION

PIZZA SALES SQL ANALYSIS | MYSQL 8.0